

CULTIVATION

The Digital Cultivation Workflow

From mother plant to harvest — every phase, every form, every sign-off.
The paper mountain, replaced by a system that schedules, captures and
files itself.

ORIGIN · by ILCO Farming · SAHPRA 22C · EU-GMP / GACP aligned · 17 June 2026 · Private & Confidential

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THE PLANT LIFECYCLE — ONE CONTINUOUS CHAIN

MOTHER

CLONE

VEG

FLOWER

HARVEST

HOW EVERY FORM NOW WORKS — THE RAIL

1 · Auto-scheduled

The form fires itself on its cadence (daily checks, temp/humidity, cleaning) — no one prints or hands out paper.

2 · Captured on a tablet

The team taps the reading/count and signs with a PIN. No re-keying, no lost sheets.

3 · Self-policing

Overdue or missing items escalate automatically; the manager only sees the exceptions.

4 · Self-filing evidence

Every completion is hash-chained and filed as EU-GMP inspection evidence — instantly audit-ready.

The shift for the team: print → hand out → chase → check → file → it appears → capture → sign. Done.

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01

Mother Room

THE GENETICS · MP1 & MP2

WHAT HAPPENS

Register the mother

Room (MP1/MP2), strain, quantity, **inception date**. Same strain can have mothers of different ages.

Lifecycle & cull

Configurable lifecycle (default **180 days**) → the system computes the **cull date** and flags overdue mothers.

Mother bay records

Foliar applications + health checks captured against each mother.

Source of every clone

Every cloning job starts from a registered mother — clean genealogy from day one.

Paper: Mother register · Mother cloning · Foliar log → **digital Mother register + auto foliar task**

Nursery Manager

SOP 3-CUL-6

auto cull-due alerts

02 Cloning · Nursery

CLONE ROOM · DAYS 1–14 ROOTING

ONE CLICK STARTS THE WHOLE JOB

Start a cloning job

Pick the mother → **strain + batch # fill themselves**. Enter clones, clone date, purpose (Production / New Mothers / Client).

Auto-creates ~18 tasks

14 daily Temp & Humidity (target pre-set per day) + **W1/W2/W3 mortality checks + transplant** — all dated, assigned to the nursery.

Daily clone-room forms

Cleaning schedule, daily check, activity log — auto-appear for sign-off, no paper.

Mortality & culling

Digital form — strain, count, reason, weight/batch/tag/zone → QA reviews.

Paper: New Cloning · Temp/Humidity ×14 · Cleaning · Mortality · Culling · Jiffy → **all auto-scheduled to sign off**

 **Nursery Manager PWA** — the input-control point for cloning. The NM runs the whole phase from a phone in the clone room: **start the job**, capture **daily temp/humidity + mortality**, and **sign off** on the spot. Every clone-room form routes to the NM by default.

Nursery Manager / Cultivator

SOP 3-CUL-7 / 3-CUL-9

batch # auto · W1–3 auto

NM PWA controlled

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Nursery Manager PWA

THE SOURCE OF INPUT · WHERE THE DATA IS BORN

Every number that flows through the platform starts here. The Nursery Manager works from a **phone or tablet in the clone room** — an installable app (PWA) that captures the cloning phase at the point of action. Nothing is written on paper and re-keyed later; the data is **born digital, at the source**.

WHAT THE NM CAPTURES — AND WHAT IT HYDRATES

► Start a cloning job

Mother → strain + batch # auto → generates the batch, clone tray, and the whole 18-task schedule.

► Daily Temp & Humidity

3 readings/day vs target → hydrates the environmental record + flags out-of-range.

► Mortality & culling

Count + reason + photo → feeds yield/loss analytics + QA review.

► Sign-off on the spot

PIN signature → becomes hash-chained EU-GMP evidence instantly.

ONE INPUT → THE WHOLE SYSTEM FILLS ITSELF



NM PWA

(clone room)



BayGrid & pots

Grow calendar & tasks

Compliance evidence pack

Owner 360 dashboard

Why it matters: capture-at-source = no paper, no re-keying, no lag. The data is accurate, contemporaneous and audit-ready the moment it's entered — and it populates every downstream view automatically.

Nursery Manager

PWA · phone/tablet

capture-at-source · offline-tolerant

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Who Drives, Who Executes

ONE SCHEDULE · ONE ACCOUNTABLE OWNER PER TASK

Every cultivation task has a clear chain: the **Head of Cultivation drives the plan**, the system generates the work, and the **role that performs it executes**. Nobody hands out paper; nobody guesses who's responsible.

DRIVER — Head of Cultivation

Lou (Lourens Eksteen) plans the phase in the **Grow Calendar**. Owns the **Cloning Schedule & Harvest Request**. Start a batch → the whole task set generates automatically.

EXECUTOR — Nursery Manager

The NM **executes** what the schedule generates — the daily clone-room care — on the **NM PWA** (phone). He doesn't plan; he captures-at-source.

WHAT A CLONING BATCH AUTO-GENERATES → ALL TO THE NURSERY MANAGER

Task	When	Allocated to
Daily Temp & Humidity ×14 (target pre-set per day)	Day 1–14	Nursery Manager
Mortality Check — W1	Day 7	Nursery Manager
Mortality Check — W2	Day 14	Nursery Manager
Mortality Check — W3	Day 21	Nursery Manager
Transplant	Rooting-complete	Nursery Manager

THE DAILY T&H TARGET STEPS DOWN AUTOMATICALLY

Day 1–3
20–24°C
95–100% RH

Day 4–6
24–26°C
90–95% RH

Day 7–9
24–25°C
85–90% RH

Day 10–12
23–25°C
80–85% RH

Day 13–14
22–25°C
70–75% RH

Commissioning rule: a form only generates a task when its role has a **real, named person**. No owner = no task (it stays dark until that person is added). No dummy data, ever.

Live logins: Lou (HoC · driver) lou@ilcofarming.co.za · Nursery Mgr (executor) nm@ilcofarming.co.za · Loraine (Cultivation Admin) fm@ilcofarming.co.za

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03

Vegetative

GH1 · TRANSPLANTED CLONES

TIGHTLY CONTROLLED BY STRAIN — DOWN TO THE POT

GridView drill-down

Greenhouse → Bay → Row → **Subrow (one strain, locked)**
→ 130 pots, each a permanent QR. Drill to any plant.

Strain integrity

One strain per subrow, enforced + audited — no drift. Status lights: ● active · ● flagged · ● dead.

Daily capture

Daily check · Env Log (temp/RH) · IPM scouting · Feeding — auto-appear daily for sign-off.

Grow calendar

Day-by-day schedule with the governing SOP, relevant role, and staff assignment per day.

Paper: Daily check · Env readings · IPM · Feed log → **auto-scheduled daily tasks**

Cultivator / Head of Cultivation

SOP 3-CUL-3 / 3-CUL-5

subrow strain-lock

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04 Flower → Harvest

GH2 · FLIP · 8 WEEKS · HARVEST

THE CALENDAR DRIVES THE WORK

Flip & flower schedule

Grow calendar fires each task — flip, defoliate, flush — on its day, to the assigned cultivator.

Daily monitoring

Env Log + daily checks continue; weight-variance and anomalies auto-alert senior roles.

Harvest request

Raise a harvest request → creates the batch (B-2026-NNN) and hands off to Processing.

Clean genealogy

Mother → clone tray → plant → batch — the full chain of custody, audit-ready.

Paper: Grow log · Harvest request → calendar tasks + digital harvest request → batch

Head of Cultivation

SOP 3-CUL-4 / 3-CUL-10

→ hands off to Processing

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THE RESULT — FOR THE CULTIVATION TEAM

11 / 11 forms digital

Every clone-room & cultivation form has a digital home — auto-scheduled or one-tap capture.

Nothing to print or file

No printing, handing out, chasing or filing. The system does the admin; the team does the plants.

Inspection-ready, always

Every sign-off is hash-chained EU-GMP evidence — the inspection pack builds itself.

Manager sees exceptions

Smart notifications surface only what needs attention — overdue, flagged, out-of-range.

Bottom line: the paperwork mountain becomes a short list of **review-and-sign** on a tablet. The system schedules, chases and files — the team supervises.